

Table 2: The Truth Table for the Conjunction of Two Propositions

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Note: In natural language, there is a subtle difference in meaning between "and" and "but", but here we will use them interchangeably.

Example 5: Find the conjunction of the propositions p and q where p is the proposition "Rebecca's PC has more than 16GB free hard disk space" and q is the proposition "The processor in Rebecca's PC runs faster than 1 GHz."

Soln: Rebecca's PC has more than 16GB free hard disk space, and its processor runs faster than 1 GHz.

Definition 3: Let p and q be propositions. The disjunction of p and q , denoted by $p \vee q$, is the proposition "p or q." The disjunction $p \vee q$ is false when both p and q are false, and is true otherwise.

Table 3: The Truth Table for the Disjunction of Two Propositions

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

$\vee \rightarrow$ inclusive or

Example 7: What is the disjunction of the propositions p and q , where p and q are the same propositions as in Example 5?

Soln: Rebecca's PC has more than 16GB free hard disk space, ~~and its processor~~ or the PC's processor runs faster than 1 GHz.

Definition 4: Let p and q be propositions. The **exclusive or** of p and q , denoted by $p \oplus q$ (or $p \text{ XOR } q$), is the proposition that is true when exactly one of p and q is true, and is false otherwise.